Computational Fluid Dynamics
Associate Professorship of Hydromechanics
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CFD Assignment 1

Group 8

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Contents

1 Introduction **1**

2 Mathematical Background **1**
2.1 Problem Statement 1
2.2 Analytical Solution 1

3 Assignment 1a: Finite Difference Method **2**
3.1 Task Overview 2
3.2 Numerical Implementation 2
3.3 Test Cases and Results 3
3.4 Convergence Study 5

4 Assignment 1b: Finite Volume Method **6**
4.1 Task Overview 6
4.2 Numerical Implementation 6
4.3 Test Cases and Results 8
4.4 Error and Accuracy Analysis 9

5 Conclusion **10**

1 Introduction

In this project, we work on solving a well-known 1-D steady-state convection-diffusion equation using numerical methods. This equation is often seen in fluid dynamics and heat transfer and represents the balance between how stuff moves and how it spreads out.

We tested two common methods: the finite difference method (FDM), which we did in Assignment 1a, and the finite volume method (FVM), which we explored in Assignment 1b. In both cases, we applied Dirichlet boundary conditions and tried different flow directions to see how the solution behaves.

2 Mathematical Background

2.1 Problem Statement

We're solving this steady 1-D convection-diffusion equation:

$$-U_0 \frac{d\phi}{dx} + \Gamma \frac{d^2\phi}{dx^2} = 0, \quad x \in [0, 2\pi],$$

with Dirichlet boundary conditions:

$$\phi(0) = 0, \quad \phi(2\pi) = 1.$$

2.2 Analytical Solution

The exact solution is:

$$\phi_{\text{ex}}(x) = \frac{e^{U_0 x/\Gamma} - 1}{e^{2\pi U_0/\Gamma} - 1}.$$

We use this to check how good our numerical solutions are. Specifically, we calculate the relative error at $x = \pi$:

$$E_{\text{rel}} = \frac{|\phi_{\text{num}}(\pi) - \phi_{\text{ex}}(\pi)|}{\phi_{\text{ex}}(\pi)}.$$

This helps us evaluate how accurate each method is at different grid sizes.

3 Assignment 1a: Finite Difference Method

3.1 Task Overview

We used the finite difference method to solve the same equation, testing both upwind and central schemes for the convective term, and comparing their performance under different grid sizes.

3.2 Numerical Implementation

The method was implemented in MATLAB. The convective term used either upwind or central differencing, while the diffusive term always used central differencing. Dirichlet boundary conditions were applied directly, and the resulting system was solved.

Upwind Scheme The first derivative is approximated as:

$$\frac{d\phi}{dx} \approx \frac{\phi_i - \phi_{i-1}}{\Delta x}$$

with coefficients:

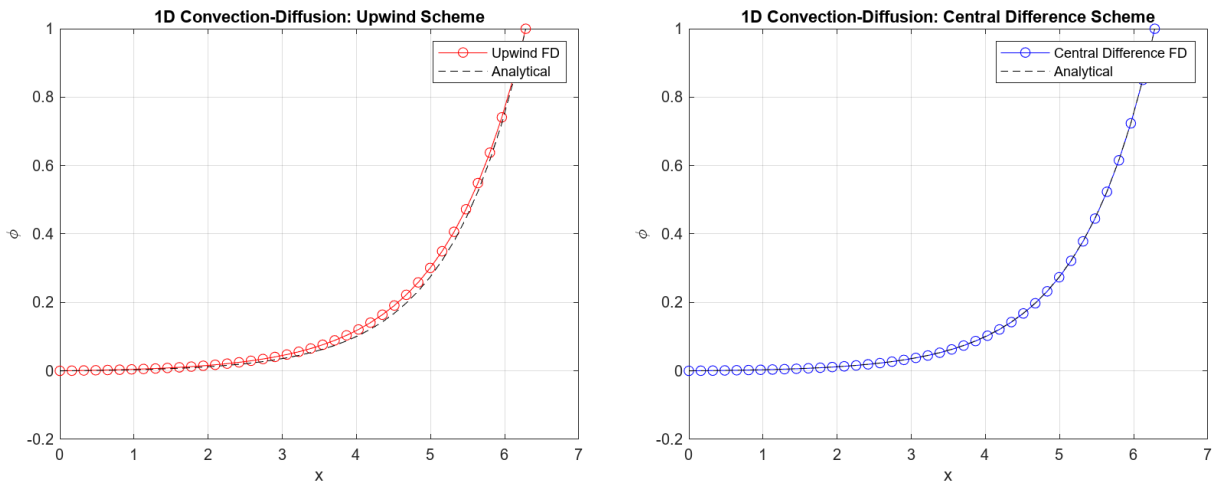
$$a_W = \frac{U_0}{\Delta x} + \frac{\Gamma}{\Delta x^2}, \quad a_P = -\frac{U_0}{\Delta x} - \frac{2\Gamma}{\Delta x^2}, \quad a_E = \frac{\Gamma}{\Delta x^2}.$$

Central Difference Scheme The first derivative is approximated as:

$$\frac{d\phi}{dx} \approx \frac{\phi_{i+1} - \phi_{i-1}}{2\Delta x}$$

with coefficients:

$$a_W = -\frac{U_0}{2\Delta x} + \frac{\Gamma}{\Delta x^2}, \quad a_P = -\frac{2\Gamma}{\Delta x^2}, \quad a_E = \frac{U_0}{2\Delta x} + \frac{\Gamma}{\Delta x^2}.$$



(a) 1-D Upwind Convection

(b) 1-D Central Convection

Figure 1: Numerical vs. Analytical Solutions for All Test Cases

Figure 1 shows how both schemes compare with the analytical solution. The upwind scheme is known to be more stable in convection-dominated cases, but it tends to smooth out sharp features in the solution. The central scheme is more accurate when the grid is fine enough, but it can show oscillations when the resolution is too low. (In Figure 1, both methods behave well due to sufficient resolution, so their contrasting behaviors are not yet clearly visible.) We'll take a closer look at this in the next section.

3.3 Test Cases and Results

We tested these four cases to see how the methods behave:

Case	U_0	Γ	Grid Points
1	-10.0	1.0	5
2	+10.0	1.0	5
3	-10.0	1.0	51
4	+10.0	1.0	51

Table 1: Test cases for both upwind and central difference schemes

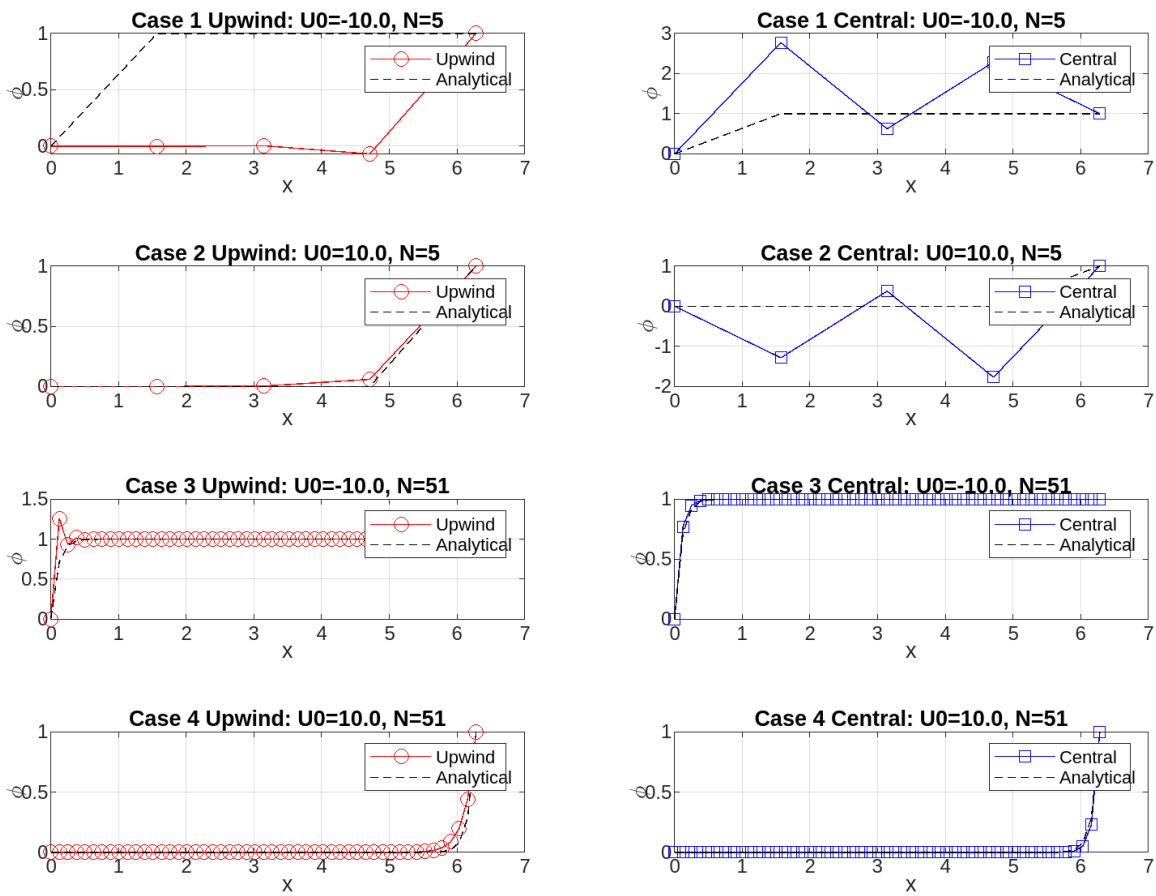


Figure 2: Numerical vs. Analytical solutions for different test cases using upwind and central schemes.

To better show the results, we let MATLAB auto-scale each plot's y-axis. This made it easier to spot oscillations, though a bit harder to compare them side-by-side. On the other hand, the upwind scheme stays stable in all cases, but it clearly smooths out the profile.

In Case 1 ($U_0 < 0$) of the upwing scheme, the reversed flow direction (right to left) causes the numerical solution to appear flipped relative to the analytical one. With a coarse grid, the upwind scheme retains near-zero values across most of the domain and only reflects the boundary condition near the outlet.

In coarse grids, the central scheme shows strong oscillations, especially when convection is strong compared to diffusion (i.e., high Peclet number). This happens because the scheme is symmetric and doesn't add any damping, so it struggles when the grid isn't fine enough.

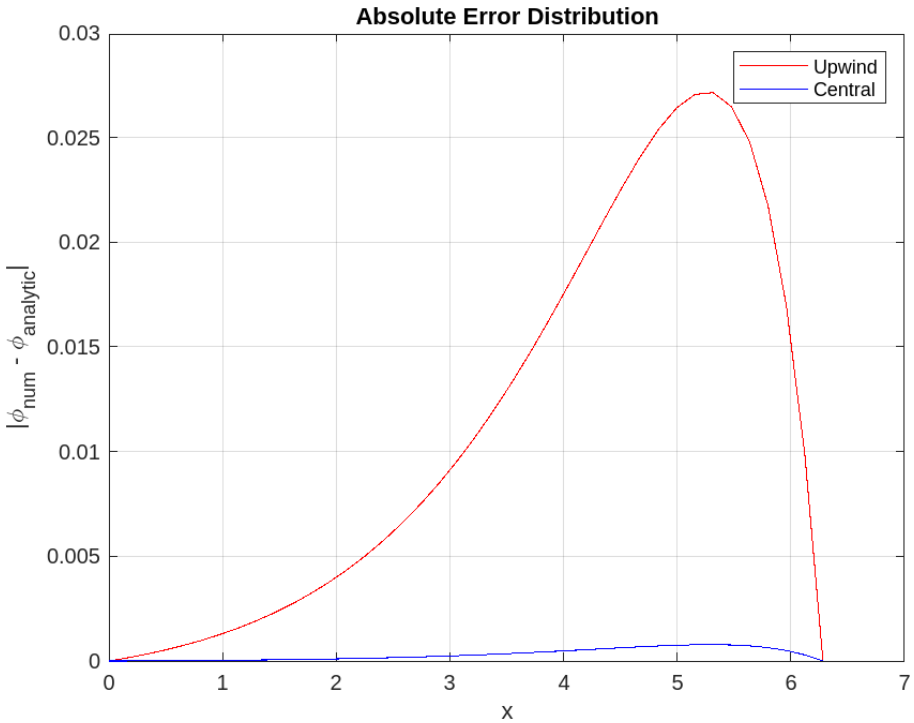


Figure 3: Absolute error between numerical and analytical solutions for upwind and central schemes at $N = 40$.

Note on Central Scheme Oscillations In Case 1 and Case 2, we observed clear oscillations in the central scheme results. This occurs when $Pe = \frac{U_0 \Delta x}{\Gamma} \gg 1$, meaning the problem is dominated by convection. Under these conditions, the discretized system starts to behave like a wave equation. If the grid resolution is too coarse, the method cannot accurately capture the solution's shape, resulting in non-physical oscillations. This highlights the advantage of the upwind scheme: although it introduces artificial diffusion, it remains stable even when the flow is strongly convection-driven.

3.4 Convergence Study

To evaluate accuracy, we ran simulations with different grid sizes and tracked the relative error at $x = \pi$. The grid sizes we tested were:

$$N = \{11,\ 21,\ 41,\ 81,\ 161\}$$

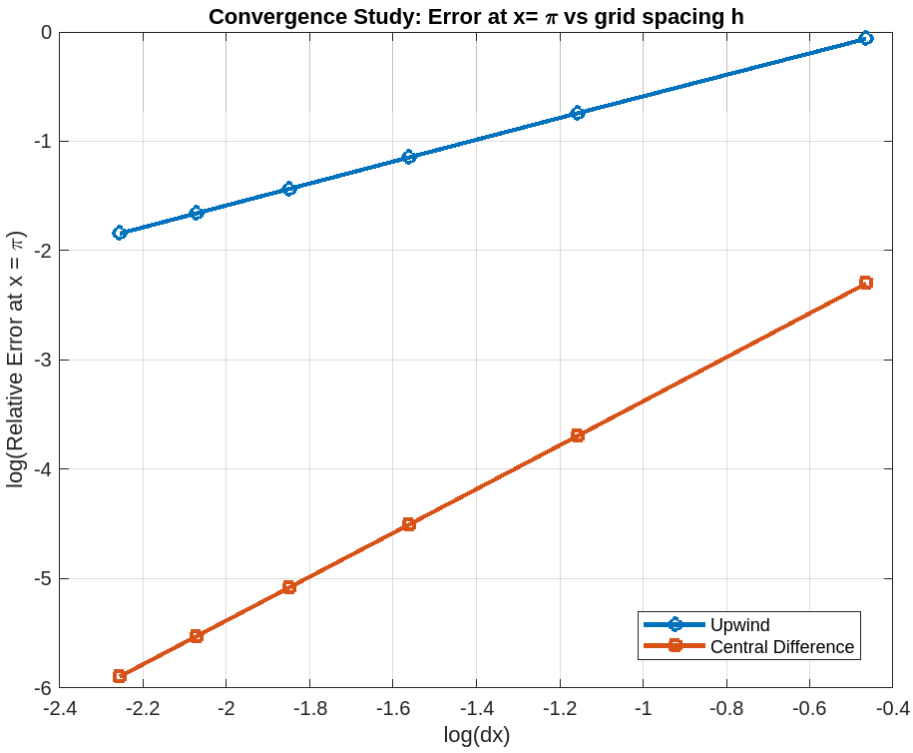


Figure 4: Convergence of upwind and central schemes: log-log plot of relative error at $x = \pi$ vs. grid spacing h.

Connection to Theory

From theory (via Taylor expansion), we know the upwind scheme gives first-order error ($\mathcal{O}(\Delta x)$), and the central scheme gives second-order error ($\mathcal{O}(\Delta x^2)$). The slopes in Figure 4 reflect this: upwind has a linear trend, while central follows a quadratic one when the grid is fine enough. So the methods are behaving as expected.

4 Assignment 1b: Finite Volume Method

4.1 Task Overview

In Assignment 1b, we switched to the finite volume method (FVM) to solve the same problem. For the convective term, we still used central differencing. For boundaries, we applied one-sided approximations. Compared to FDM, FVM tends to be more stable, especially when convection is strong.

4.2 Numerical Implementation

In FVM, each value ϕ_P is stored at the center of a control volume. The neighbors to the left and right are ϕ_W and ϕ_E, and we approximate the values at the faces (w and e) using linear interpolation.

We set up a uniform grid with 51 control volumes and used $U_0 = 10$, $\Gamma = 1$. The governing equation was rewritten in integral form and turned into a system of equations. Central differencing was used for the convective term, and boundary faces were handled using one-sided logic.

Boundary Treatment for the Rightmost Control Volume In the finite volume method, boundary conditions are applied at the faces of the boundary control volumes. At the right boundary, the Dirichlet condition is given as $\phi_R = 1$, but since this value lies at the east face of the last control volume, it must be incorporated indirectly. The resulting stencil is therefore referred to as "incomplete" and requires special treatment.
We begin from the integral form of the governing equation over the last control volume:

$$\phi_w U_0 S - \phi_e U_0 S - \Gamma \left(\frac{\partial \phi}{\partial x}\right)_w S + \Gamma \left(\frac{\partial \phi}{\partial x}\right)_e S = 0$$

The face values and gradients are approximated as follows:

$$\phi_e = \frac{\phi_P + \phi_R}{2}, \quad \phi_w = \frac{\phi_W + \phi_P}{2}$$

$$\left(\frac{\partial \phi}{\partial x}\right)_e = \frac{\phi_R - \phi_P}{\Delta x/2}, \quad \left(\frac{\partial \phi}{\partial x}\right)_w = \frac{\phi_P - \phi_W}{\Delta x}$$

Substituting into the balance equation:

$$\left(\frac{\phi_W + \phi_P}{2}\right) U_0 - \left(\frac{\phi_P + \phi_R}{2}\right) U_0 - \Gamma \left(\frac{\phi_P - \phi_W}{\Delta x}\right) + \Gamma \left(\frac{\phi_R - \phi_P}{\frac{\Delta x}{2}}\right) = 0$$

Simplifying:

$$-\left(\frac{3\Gamma}{\Delta x}\right) \phi_P + \left(\frac{\Gamma}{\Delta x} + \frac{U_0}{2}\right) \phi_W = \left(\frac{U_0}{2} - \frac{2\Gamma}{\Delta x}\right) \phi_R$$

Rearranged into matrix form, this gives the coefficients for the last row in the system: Given the boundary condition $\phi_R = 1$, the final row of the matrix and right-hand side vector becomes:

$$A(\text{cells}, \text{cells}) = -\frac{3\Gamma}{\Delta x}$$
$$A(\text{cells}, \text{cells} - 1) = \frac{\Gamma}{\Delta x} + \frac{U_0}{2}$$
$$b(\text{cells}) = \left(\frac{U_0}{2} - \frac{2\Gamma}{\Delta x}\right)\phi_R \quad (\text{with } \phi_R = 1)$$

This completes the special treatment of the right boundary using the given Dirichlet condition at the cell face.

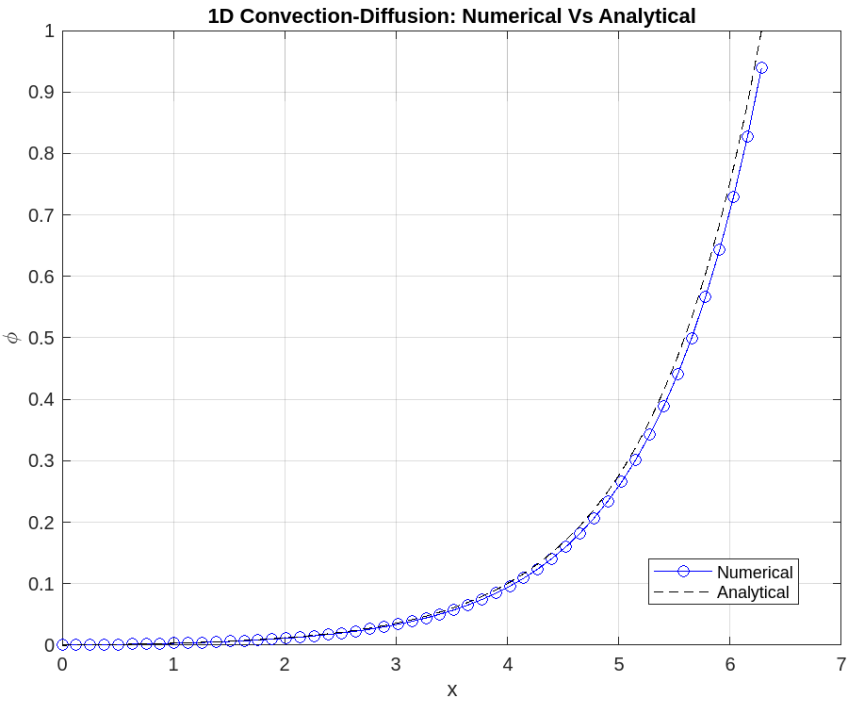


Figure 5: Numerical solution vs analytical result using finite volume method.

Figure 5 shows that the FVM solution follows the analytical one very closely—even though the flow is dominated by convection. This supports what we saw in the lecture: FVM's conservative formulation (based on flux balance) helps keep the solution stable even when using central differences.

4.3 Test Cases and Results

We used the same four test cases from Assignment 1a, just to make it easier to compare:

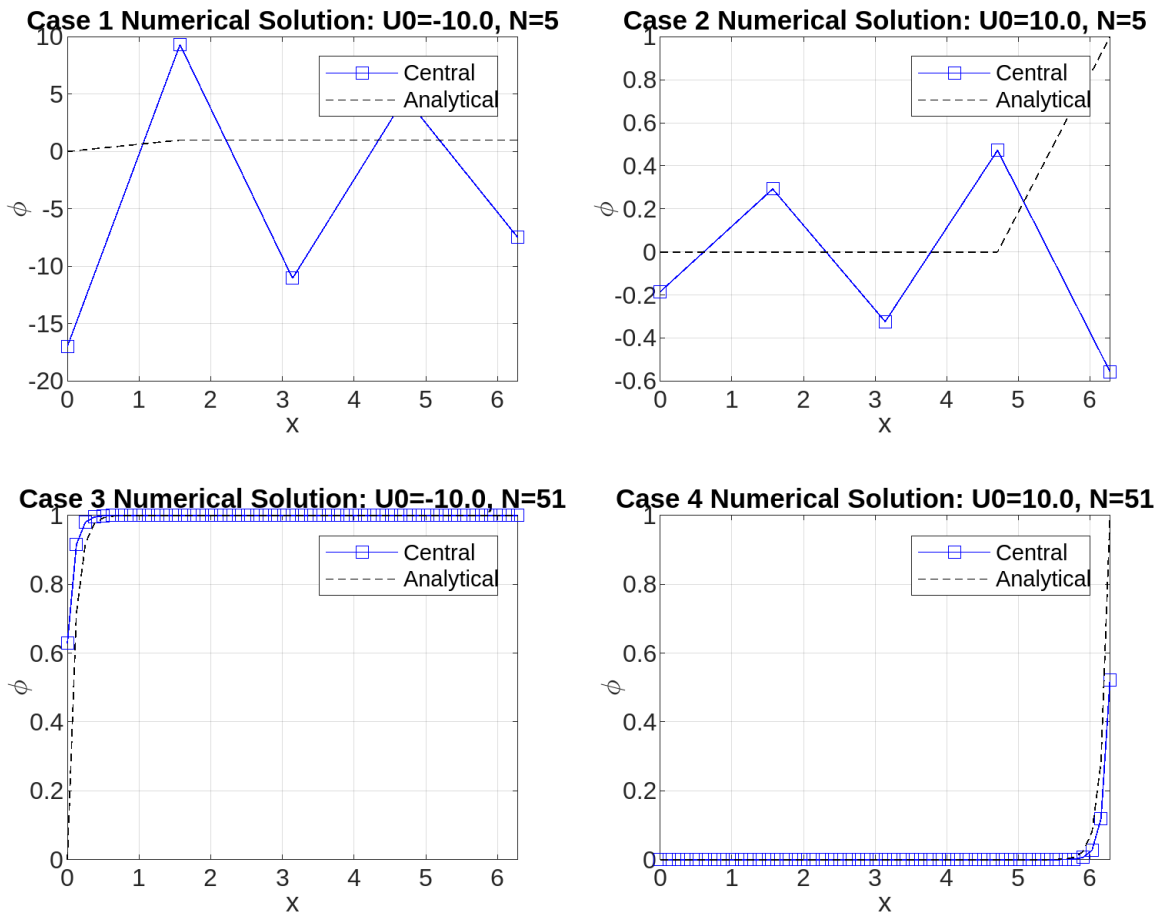
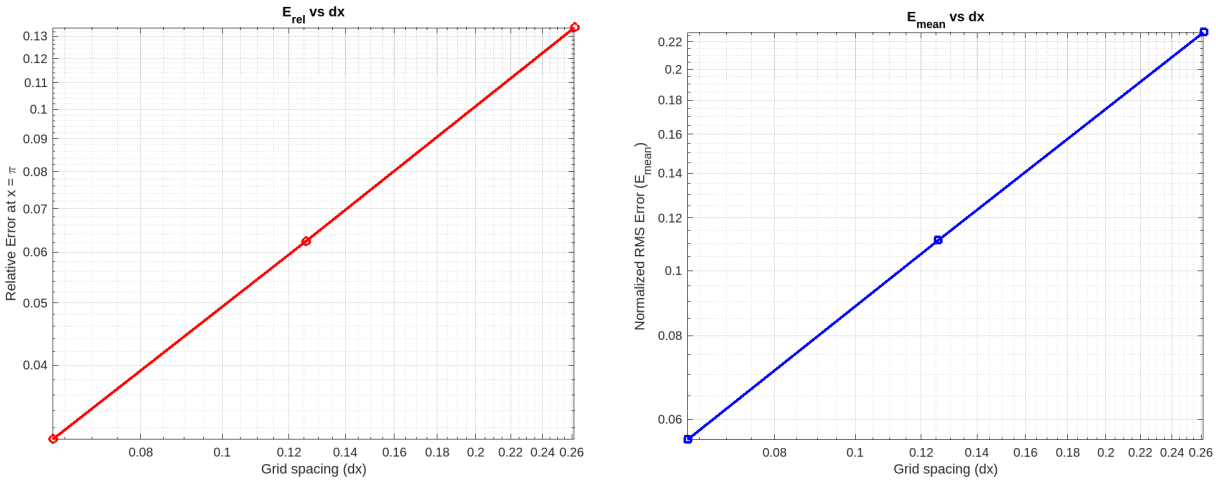


Figure 6: Numerical solution vs analytical result using finite volume method.

As shown in Figure 6, the FVM results vary depending on the flow direction and grid resolution. For coarse grids ($N = 5$), Cases 1 and 2 exhibit noticeable oscillations—particularly in Case 1 ($U_0 = -10$), where the numerical solution deviates significantly from the analytical profile. Although FVM is generally considered more stable due to its conservative flux formulation, in this case the oscillations are not necessarily less severe than those observed with the central scheme in FDM. This suggests that while FVM offers advantages in many scenarios, it still requires sufficient grid resolution to remain stable when the Peclet number is high.
When the grid is finer ($N = 51$), the solution almost perfectly matches the analytical one in all cases. So, overall, FVM handles strong convection quite well if you have decent resolution.

4.4 Error and Accuracy Analysis

To evaluate the accuracy of the finite volume method, we calculated the relative error at $x = \pi$ and the normalized RMS error over the entire domain across several grid resolutions. The number of control volumes was varied to adjust the spacing dx, and the resulting errors are shown in the log-log plots in Figure 7.



(a) Relative error E_{rel} vs dx

(b) Normalized RMS error E_{mean} vs dx

Figure 7: Log-log plot of error vs grid spacing for the finite volume method.

Connection to Theory

Since we're using central differences and linear interpolation, the truncation error is theoretically expected to be proportional to Δx^2, indicating second-order accuracy. However, as seen in Figure 7, the observed slopes in our log-log plots are lower than expected. The RMS error E_{mean} shows a slope close to 1, and the pointwise error E_{rel}, evaluated at $x = \pi$, is even flatter—closer to 0.5. This discrepancy may be due to several factors, such as the limited number of grid refinements tested and the sensitivity of pointwise error to whether π aligns exactly with a control volume center. Additionally, the normalization in the RMS error calculation might have a minor influence on the slope.

Despite these deviations, both errors decrease consistently with grid refinement, confirming the correctness of the implementation and the expected convergence behavior of the finite volume method.

5 Conclusion

In this assignment, we compared two ways of solving the 1-D steady convection-diffusion equation: the finite difference method (FDM) and the finite volume method (FVM).

The upwind scheme in FDM was very stable, even when convection was strong, but it introduced noticeable numerical diffusion that smoothed out sharp features of the solution. The central difference scheme gave much more accurate results on fine grids but showed oscillations when the resolution wasn't high enough—especially when convection dominated.

The FVM, on the other hand, handled convection-dominated cases reasonably well overall. Even though it also used central differencing, its conservative structure (focusing on total flux balance) is theoretically expected to improve stability. However, this advantage was not consistently observed on coarse grids—for example, in Case 1, FVM central exhibited even stronger oscillations than FDM central. This highlights that while FVM's formulation offers potential benefits, sufficient grid resolution is still critical in convection-dominated flows.

When we looked at convergence, we found what theory predicted: the upwind scheme gives first-order accuracy, and the central scheme gives second-order accuracy—as long as the grid is fine enough. The RMS error across the domain was more consistent than the error measured at a single point, which sometimes fluctuated due to alignment with the grid.

In short, central schemes can be very accurate, but they need good resolution to avoid instability. Upwind is safer, but less precise. FVM with central differencing gave us a reasonable middle ground—it was mostly accurate and stable in our test cases, though still susceptible to oscillations when the grid was coarse. With sufficient resolution, however, it proved to be a reliable method for handling convection-dominated problems.